

Euro Area Monetary & Financial Conditions

An Empirical Analysis of Monetary-Policy Transmission

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Python, SQL, ECB and Eurostat Data

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1 Introduction

Monetary-policy decisions affect the economy through a sequence of financial and macroeconomic transmission mechanisms. Changes in central-bank policy rates influence short-term money-market rates, bank funding conditions, borrowing costs faced by households and firms, aggregate demand and, eventually, inflation.

This project provides an exploratory empirical analysis of monetary-policy transmission in the euro area from January 2015 to July 2026. It combines official data from the European Central Bank (ECB) and Eurostat in a reproducible Python and SQL workflow.

The analysis focuses on a simplified transmission chain:

ECB policy rate $\rightarrow$ €STR $\rightarrow$ bank borrowing rates $\rightarrow$ economic conditions $\rightarrow$ inflation

The main research question is:

How has ECB monetary policy been transmitted to money-market rates, bank borrowing conditions and inflation in the euro area since 2015?

The objective is descriptive rather than causal. The analysis therefore documents comovement and lag structures without interpreting simple correlations as identified monetary-policy effects.

2 Data

The analytical dataset combines monetary, financial and macroeconomic variables from official ECB and Eurostat sources.

The sample contains 139 monthly observations covering January 2015 to July 2026.

The principal variables are:

- ECB Deposit Facility Rate (DFR);
- euro short-term rate (€STR);
- cost of borrowing for households for house purchase;
- cost of borrowing for non-financial corporations;
- headline Harmonised Index of Consumer Prices (HICP) inflation;
- core HICP inflation excluding energy, food, alcohol and tobacco;
- euro-area unemployment rate.

ECB policy-rate observations and the business-day €STR series are aggregated to monthly frequency. The resulting series are combined with monthly bank lending and macroeconomic indicators.

The €STR is available only from October 2019. Observations before that date are therefore intentionally missing rather than imputed.

The data pipeline retrieves the source series programmatically, harmonises dates and frequencies, performs validation checks and constructs a single monthly analytical dataset. The processed data are also stored in a SQLite database for SQL-based exploration.

3 Monetary and Financial Conditions

Figure 1 compares the ECB Deposit Facility Rate with the €STR.

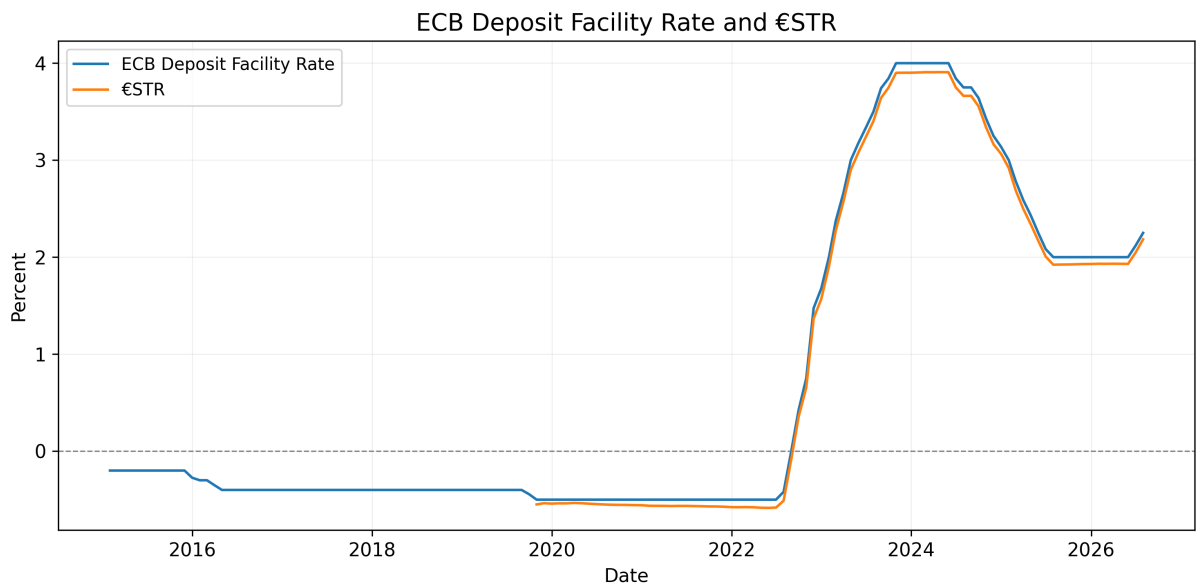


Figure 1: ECB Deposit Facility Rate and €STR

The two rates move closely together over the available €STR sample. This illustrates the strong connection between the ECB policy rate and overnight unsecured money-market conditions.

Figure 2 shows headline and core HICP inflation.

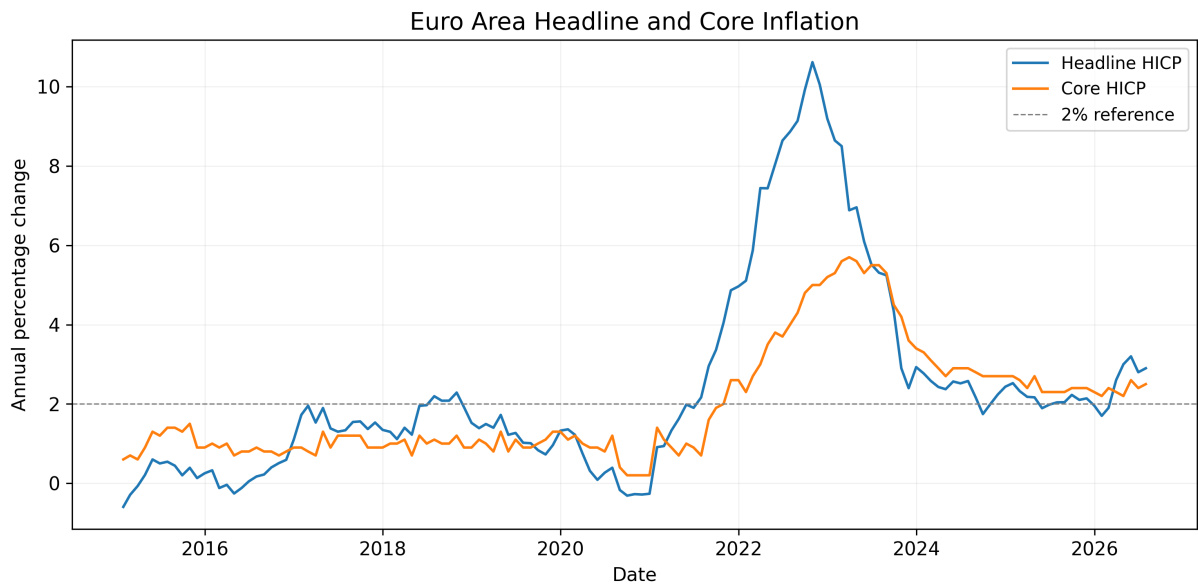


Figure 2: Euro-area headline and core inflation

The sample encompasses markedly different inflation environments, including the low-inflation period before the pandemic, the subsequent inflation surge and the later moderation. Headline inflation displays substantially greater short-run volatility than core inflation.

4 Transmission to Bank Borrowing Rates

Figure 3 compares the DFR with borrowing rates faced by households and non-financial corporations.

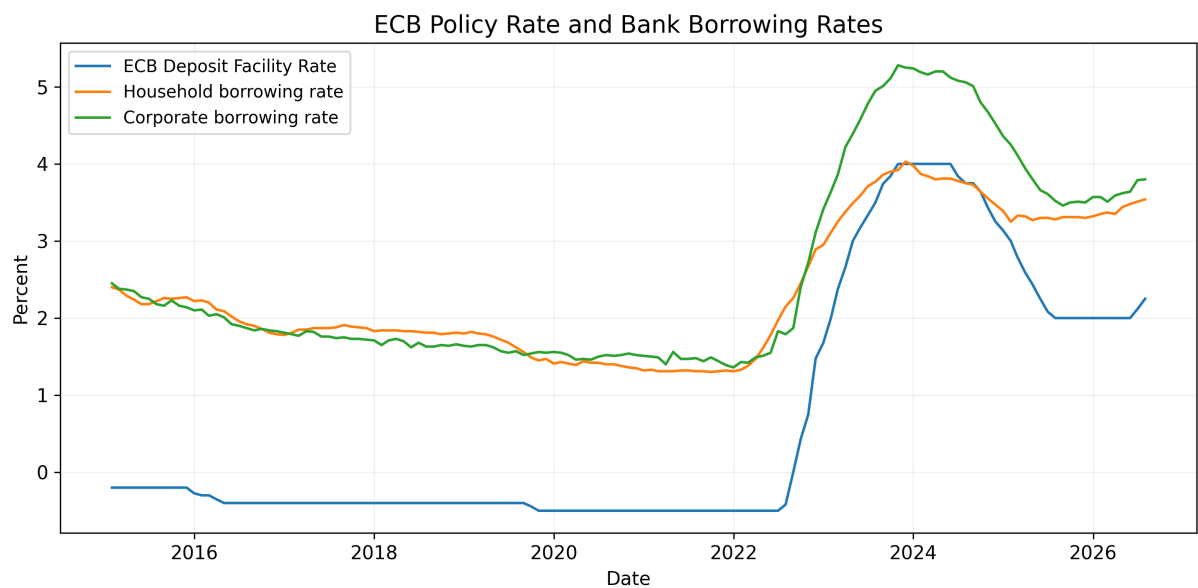


Figure 3: ECB policy rate and bank borrowing rates

Bank borrowing rates broadly move in the same direction as the policy rate, although their levels and adjustment patterns differ.

To examine this relationship without relying exclusively on highly persistent interest-rate levels, the analysis considers monthly changes:

$$\Delta r_t = r_t - r_{t-1}.$$

For each lag k , the descriptive statistic is

$$\rho_k = \text{Corr} \left(\Delta r_t^{bank}, \Delta DFR_{t-k} \right).$$

Figure 4 reports these correlations for lags from zero to twelve months.

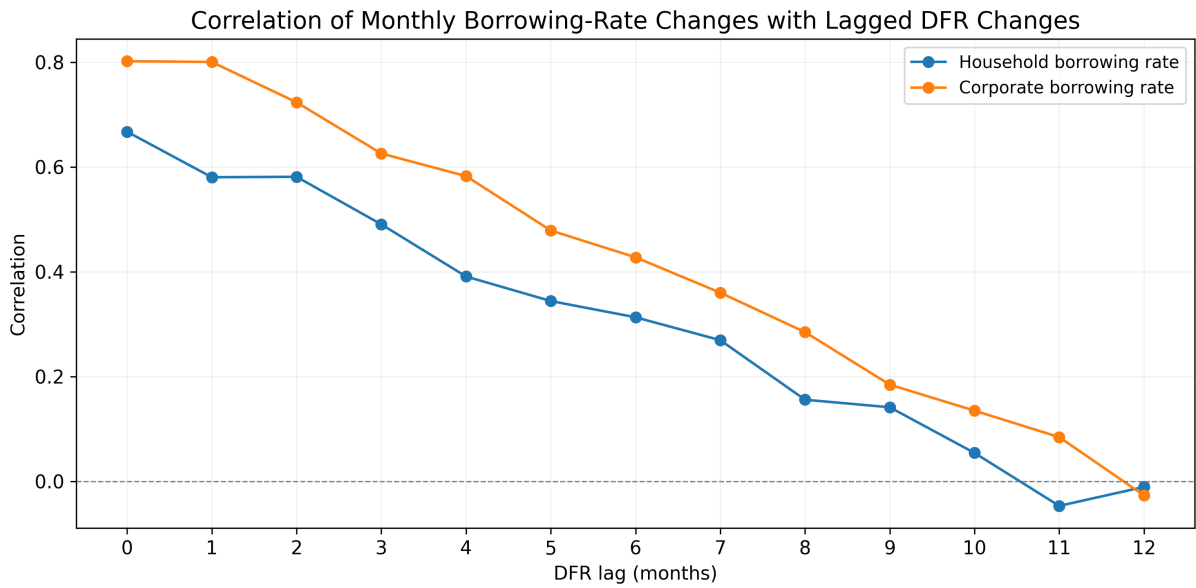


Figure 4: Correlation of monthly borrowing-rate changes with lagged DFR changes

The association is strongest contemporaneously and at short lags. At lag zero, the correlation is approximately 0.67 for household borrowing rates and 0.80 for corporate borrowing rates.

The correlations subsequently decline as the DFR lag increases. Corporate borrowing rates display a particularly strong association with current and recent policy-rate changes in the full sample.

These results are consistent with relatively rapid pass-through from policy rates to bank lending conditions, but they should not be interpreted as causal estimates or as identifying a precise transmission lag.

5 Borrowing-Rate Spreads

For descriptive purposes, borrowing-rate spreads relative to the DFR are defined as

$$Spread_t = r_t^{bank} - DFR_t.$$

Figure 5 plots these differences.

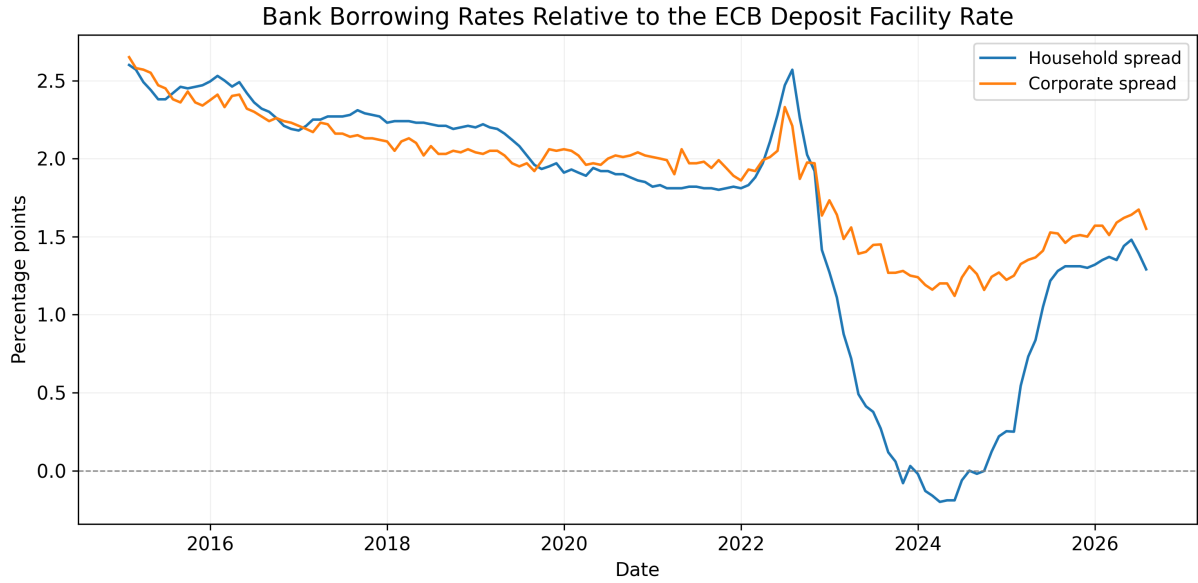


Figure 5: Bank borrowing rates relative to the ECB Deposit Facility Rate

These spreads should not be interpreted as bank profit margins. Lending rates reflect differences in maturity, credit risk, funding costs, operating costs, loan composition and competitive conditions, among other factors.

The figure instead provides a simple descriptive measure of how borrowing rates evolve relative to the policy rate across different monetary-policy environments.

6 Monetary Policy and Inflation

The relationship between monetary policy and inflation is more difficult to interpret because policy itself responds to inflation and broader economic conditions.

To reduce the influence of persistence in the levels of both series, the analysis examines correlations between monthly changes in inflation and lagged monthly changes in the DFR:

$$\rho_k = \text{Corr}(\Delta\pi_t, \Delta DFR_{t-k}).$$

Figure 6 reports the resulting correlations.

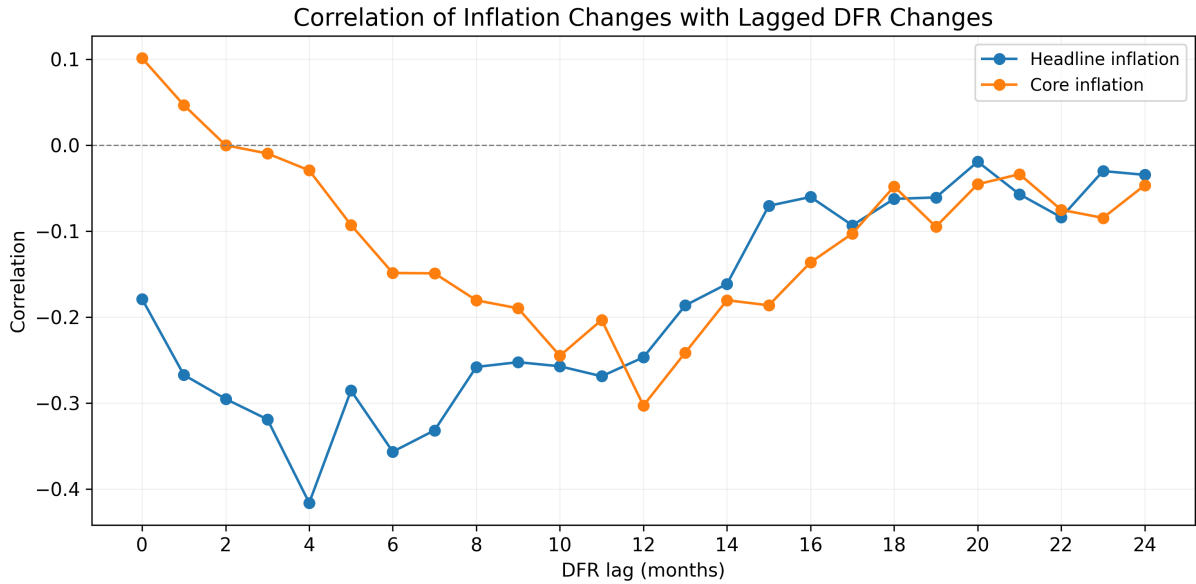


Figure 6: Correlation of inflation changes with lagged DFR changes

For headline inflation, the strongest negative correlation in the sample occurs at approximately four months, at around -0.42 .

For core inflation, the strongest negative correlation occurs at a longer lag of approximately twelve months, at around -0.30 .

The difference is consistent with headline inflation being more volatile and responsive to short-run price movements, while core inflation is typically more persistent.

However, these correlations cannot establish that a DFR change causes an inflation change after a particular number of months. ECB policy decisions respond endogenously to inflation and economic conditions, and both policy rates and inflation may respond to common macroeconomic shocks.

7 Robustness

A simple robustness exercise evaluates whether the observed relationship between policy-rate changes and bank borrowing-rate changes is concentrated in the unusually large monetary-policy adjustments of 2022–2024.

The lagged correlations are therefore recalculated after excluding observations from 2022 through 2024.

For household borrowing rates, the contemporaneous correlation falls from approximately 0.67 in the full sample to approximately 0.12 outside 2022–2024.

For corporate borrowing rates, the corresponding correlation falls from approximately 0.80 to approximately 0.57 and remains positive over the first several lags.

This exercise suggests that the strong full-sample relationship for household borrowing rates is substantially influenced by the 2022–2024 period. The relationship for corporate borrowing rates appears more robust across periods, although it is also weaker outside the excluded interval.

These results reinforce the importance of interpreting simple full-sample correlations in the context of changing monetary-policy regimes.

8 Limitations

This analysis is exploratory and does not identify exogenous monetary-policy shocks.

The ECB adjusts policy rates in response to inflation, economic activity, financial conditions and expectations. Consequently, correlations between policy rates and subsequent macroeconomic variables combine monetary-policy transmission with endogenous policy responses and common shocks.

The monthly aggregation of financial variables may also obscure higher-frequency adjustment around individual policy decisions. Bank lending-rate series differ in composition and maturity from the DFR and therefore cannot be interpreted as direct mark-ups over the policy rate.

More advanced extensions could use high-frequency identified monetary-policy shocks, distributed-lag regressions, local projections or country-level panel data.

9 Conclusion

This project documents several features of euro-area monetary-policy transmission since 2015.

First, the €STR closely tracks the ECB Deposit Facility Rate, illustrating strong transmission from the policy rate to overnight money-market conditions.

Second, monthly changes in bank borrowing rates are strongly associated with current and recent changes in the DFR, particularly for corporate borrowing rates.

Third, correlations between policy-rate changes and inflation changes display longer and less straightforward lag patterns. Headline inflation exhibits its strongest negative correlation at a shorter lag than core inflation in this sample.

Finally, the robustness exercise demonstrates that the strength of estimated relationships can depend materially on the monetary-policy period considered.

Taken together, the results illustrate both the observable transmission of monetary policy through financial conditions and the limitations of using simple correlations to infer macroeconomic causal effects.

Technical Implementation

The complete reproducible workflow is implemented using Python, pandas, matplotlib, SQL and SQLite.

Official data are retrieved programmatically from ECB and Eurostat sources. The repository contains the data-download and cleaning pipeline, SQL analysis, Python analytical code, generated figures and intermediate analytical results.